

Phantom Mass: An Intrinsic Per-Element Measure of Abstract-Domain Conflation, with Registered Experiments on Exactly When It Predicts Analysis Error

Chris Brink falsework.dev **Version.** Draft v0.3, September 2026 — DOI [10.5281/zenodo.22802676](https://doi.org/10.5281/zenodo.22802676); not yet submitted to a venue. (v0.3, after a literature-positioning review: the note no longer claims the *count* as a contribution in any form — $s(U)$ is verified here to coincide both with the CDG quasi-metric at (U, qU) and with the model-counting false-positive measure of Zheng–Yao (arXiv 2606.21992, June 2026), and §1 now positions against the three existing measurement families — quasi-metric, pre-metric (Campion–Urban–Dalla Preda–Giacobazzi, SAS 2023), and model counting — reserving the claim to the *direction of use*: intrinsic and prior, with a theory of when the predictive channel exists and a proof of when it does not. v0.2, same day as v0.1, after referee-style review: the confusion-class lemma restated and kernel-checked at `ClosureOperator` generality, closing a prose-weld where the uco claim cited the nucleus lemma; the “not forced” status of Theorem 6.1 made kernel-grade via `step_not_meet_preserving`; the partition mechanism promoted to Proposition 4.1 with the abstract-side/concrete-side grades separated; all headline numbers to three decimals with the Phase 1 margin of 0.003 flagged as prominently as the Phase 2 margin of 0.006; abstract shortened, the map first.) **Status discipline.** Every claim carries one of five grades: **[K]** kernel-checked in Lean 4 against Mathlib4 (no `sorry`, no `native_decide`, axiom audits at most `propext`, `Classical.choice`, `Quot.sound`); **[C]** classical mathematics, cited; **[R]** empirical result from a pre-registered protocol (spec and kill conditions committed before code, dated postscripts binding, amendments logged with their nature); **[A]** structural analogy, argued not proved; **[O]** open. The grades are load-bearing: nothing below claims more than its tag.

Abstract

Does an abstract domain carry, at individual properties and with no program in sight, a number that predicts where run-level analysis error will accumulate? We test one candidate — the **phantom mass** of U under a closure operator q , the size of the conflation interval $[U, qU]$ — in three pre-registered protocols with kill conditions committed before code (31,040 exact runs on finite toy worlds). The answer is a map: **yes** for gap-filling closures — the interval domain predicts accumulated completeness error at pooled Spearman +0.503, robust to a world, seed, and boundary-convention change (+0.538, +0.629), with the input-only variant at +0.463 **[R]**; **no** for partition closures — four domains show $|q| \leq 0.117$, with a four-line mechanism (the abstract run factors through the met-class profile and is blind to within-class cardinality; Proposition 4.1) and a registered kill that closed the side after the corrected functional also failed **[R]**; **attenuated** under widening — pooled +0.306 against a registered kill line of 0.300, reported at that margin, with a registered autopsy isolating the mechanism: near-full signal where widening is quiet (+0.457), swamped where it fires (+0.102), and a quarter of error-bearing loop runs carrying zero intrinsic phantom, per the pre-registered A-W anchor **[R]**. The instrument’s license is a two-line confusion-

class lemma at closure-operator generality — meet-preservation unused — kernel-checked as `ClosureOperator.le_apply_iff` [K]. The count itself is claimed by nobody here: $s(U) = |\mathcal{Q}U \setminus U|$ is exactly the counting-measure quasi-metric of Campion, Dalla Preda and Giacobazzi (POPL 2022) instantiated at $(U, \mathcal{Q}U)$, and coincides with the model-counting false-positive measure of Zheng and Yao (2026) at $\mathcal{Q} = \gamma \circ \alpha$ [C]. What is claimed is the direction of use — intrinsic and prior rather than extrinsic and post-hoc — together with the map of when that direction works. One theorem accompanies the experiments: observers compatible with a single-edge order extension are closed under support union — reduced-product-style completeness preservation in a setting where the standard sufficient condition provably fails, the step being non-meet-preserving by a kernel-checked witness [K]. In one line: *the intrinsic conflation count predicts accumulated analysis error where the abstraction fills gaps, cannot where it forgets multiplicity, and degrades with a measured mechanism where the analysis manufactures its own imprecision.*

1. Introduction

Abstract interpretation has a graded theory of *how wrong* an analysis can be. Completeness — the case where abstraction loses nothing relative to the property observed — was characterized by Giacobazzi, Ranzato and Scozzari (JACM 2000) as a property of the abstract domain alone, refinable by domain transformers; Bruni, Giacobazzi, Gori and Ranzato (LICS 2021) localized it to derivations; Campion, Dalla Preda and Giacobazzi (POPL 2022) made it quantitative, bounding the *partial completeness* ε of a run under a supplied quasi-metric. In that framework the error is a property of a triple: program, input, domain.

This note asks a question one level below the run: does the domain itself, evaluated at individual properties with no program in sight, carry a number that *predicts* where run-level error will accumulate? The candidate number is the size of the conflation interval $[U, \mathcal{Q}U]$ — how much the closure manufactures at U — which we call phantom mass. It arose in a different setting (a program studying observer-relative structure in Heyting algebras; §6 gives the two-sentence version), and the confusion-class lemma that justifies the name needs only inflationarity, monotonicity and idempotence, so it applies to every standard abstract domain as-is: it is a closure-operator-level instrument, not a nucleus-level one [K].

Positioning, stated before the contribution. Measuring abstract-interpretation imprecision quantitatively is occupied territory, and this note claims none of it. Three families exist. The quasi-metric family: Campion, Dalla Preda and Giacobazzi (POPL 2022) supply a distance on the abstract domain and bound the ε of a run. The pre-metric family: Campion, Urban, Dalla Preda and Giacobazzi (SAS 2023) relax the quasi-metric to a pre-metric on any pre-ordered domain, covering forward completeness and non-Galois abstractions, and are explicit that the distance can be designed to whatever level of approximation one wants measured. The model-counting family: Zheng and Yao (2026) count the false-positive concrete states $|\gamma(\alpha(\varphi)) \setminus \varphi|$ of an abstraction by model counting on logical encodings, giving a client-independent precision report for real domains (Interval, Zone, Octagon, KnownBits). The exponent this note works with is not new against any of them — $s(U) = |\mathcal{Q}U \setminus U|$ is the POPL 2022 counting quasi-metric instantiated at $(U, \mathcal{Q}U)$, and it is the

Zheng–Yao false-positive count at $q = \gamma \circ \alpha$, a coincidence we state rather than bury. All three families, however, run in the same direction: given an analysis output or an abstraction of a known semantics, they measure the discrepancy that *has occurred*. Extrinsic, post-hoc.

Phantom mass is the same count pointed the other way: *intrinsic and prior* — a per-element structural coordinate of the closure, defined before any program exists, evaluated along states a program then actually visits. The question is whether that prior quantity *predicts* the realized discrepancy the other frameworks measure. If it predicted nothing, it would be a definition with good manners and no job. Whether it predicts is an empirical question, so we registered it as one — spec, thresholds, and kill conditions committed before code, in three phases, with the kill line written first: *if the intrinsic count and the run-level error decouple on the worked example, the instrument is a definition, not a predictor, and the bet ends there*. And where it cannot predict, the note aims for a proof rather than a scatter plot: the partition result (Proposition 4.1) is an information-channel impossibility — the abstract trajectory is a function of a profile that phantom mass varies independently of — which *predicted* the empirical null rather than explaining it afterward.

The three phases produced a map rather than a verdict, and the map is the contribution:

- **Where it works:** gap-filling closures (the interval domain), robustly across worlds, seeds, and boundary conventions — and even in input-only form, which needs no trajectory at all (§3).
- **Where it does not work, half by proof:** partition closures — the abstract run retains which classes are met and discards multiplicity, so cardinality-style phantom information has no channel through the abstract computation (Proposition 4.1); the absence of a concrete-side rescue is the registered empirical half (§4).
- **Where it degrades, and why:** loops under widening, where the analysis manufactures imprecision that no intrinsic quantity can see; the degradation is measured, its mechanism isolated by a registered autopsy, and the surviving signal localized to the widening-quiet region (§5).

Everything empirical here is on finite toy worlds, computed exactly. The distance from these worlds to a real analyzer is stated in §7 and not minimized. What the toys establish is the *existence and shape of the phenomenon*: the intrinsic/extrinsic gap is bridgeable for one large class of domains, unbridgeable for another, and the reason in both cases is structural, not statistical.

2. The instrument

Setting [C]. A concrete domain C (here: the powerset of a finite state space Σ , ordered by inclusion). An abstract domain in the standard uco presentation is an upper closure operator $q : C \rightarrow C$ — monotone, inflationary ($U \subseteq qU$), idempotent — whose fixpoints are the expressible properties (Cousot & Cousot 1977, 1979; the uco view of domains is standard since Giacobazzi–Ranzato–Scozzari 2000).

Definition 2.1 (phantom mass) [K]. For a closure operator q and element U , the phantom mass of U under q is $|[U, qU]|$ — the cardinality of the interval between U and its closure.

Lemma 2.2 (confusion class) [K] (`ClosureOperator.le_apply_iff`). For $U \leq X$: $X \leq qU$ iff $qX = qU$. The interval $[U, qU]$ is exactly the set of properties q cannot distinguish from U from above. Two lines from inflationary + monotone + idempotent, kernel-checked over Mathlib’s `ClosureOperator` on an arbitrary partial order — meet-preservation (the nucleus axiom) is not a hypothesis, which is what licenses applying the instrument to arbitrary ucos. (The nucleus-flavored statement this program originally used, `IsNucleus.le_apply_iff`, is the special case and sits beside it in the same file; the closure-operator form was added when a draft of this note was caught citing the nucleus lemma for the uco claim — the kind of prose-weld between artifacts this program’s method exists to catch, recorded here rather than silently repaired.)

The working coordinate. On a powerset carrier, $|[U, qU]| = 2^{\{s(U)\}}$ with $s(U) = |qU \setminus U|$: the number of spurious concrete states the closure manufactures at U . s is the informative exponent and the additive form. As a functional it is not new twice over: it is exactly the counting-measure quasi-metric $\delta(U, qU)$ of Campion–Dalla Preda–Giacobazzi instantiated at the pair (U, qU) , and it is exactly the false-positive count $|\gamma(\alpha(U)) \setminus U|$ that Zheng–Yao’s MCAI computes by model counting when the closure is $\gamma \circ \alpha$. Both bridges are stated once and relied on throughout; what this note adds to the count is its semantics (Lemma 2.2: the interval $[U, qU]$ is the closure’s confusion class at U , so s is a *local conflation* measure, not a distance-to-output) and its role (predictor evaluated on concrete trajectories, not report card on abstract results).

The two run-level quantities (targets, not instruments). For a program $p = f_n \circ \dots \circ f_1$ executed abstractly by stepwise best correct approximations ($a_0 = qU$, $a_{k+1} = q(f_{k+1}(a_k))$, the composition real analyzers use), against the concrete trajectory $V_0 = U$, $V_{k+1} = f_{k+1}(V_k)$:

- **completeness error** $\varepsilon(p, U, q) = |a_n \setminus q(V_n)|$ — spurious mass beyond the best the domain could represent; the CDG-style target;
- **the intrinsic predictor** $P(p, U, q) = \sum_k s(V_k)$ over the concrete trajectory — per-element counts at the states the program actually visits, no abstract run consulted. The input-only variant uses $s(U)$ alone.

Two free facts, registered as anchors so the experiments could not discover them. T0: if $s(V_k) = 0$ for all k , then $\varepsilon = 0$ (exact trajectory forces a complete run; proof: $a_0 = qU = U$ and $a_{k+1} = q(f_{k+1}(a_k)) = q(V_{k+1}) = V_{k+1}$ by induction). A1: phantom does *not* imply error — a class-respecting operation on a partition domain has $\varepsilon = 0$ at any phantom. Both are proved inline here and asserted as validation in every registered run [R]; neither is claimed as a finding. The statistical question lives strictly between these two poles.

3. Phase 1–1b: the gap-filling case, and its robustness

Protocol (registered before code; `absint-dictionary/PHASE1-SPEC.md`, `PHASE1B-SPEC.md`) [R]. World A: $\Sigma = \{-8..8\}$, five ucos — Parity, Mod3, Sign, Sign×Parity (partition closures), Interval ($qU = [\min U, \max U]$; the gap-filling closure) — 50 programs over a 10-operation alphabet (saturating arithmetic and guards), 48 inputs, 12,000 exact runs. Registered thresholds: within-domain pooled Spearman(P, ε) ≥ 0.5 with length-stratified median ≥ 0.3 (the stratification guard against the length

confound: longer programs accumulate both phantom and error). Kill KP1: pooled < 0.3 in every domain.

Result. The kill did not fire, and the registered universal claim failed — the split is the finding:

domain	pooled $q(P, \epsilon)$	length-stratified median	input-only $q(s(U), \epsilon)$
Parity	+0.029	-0.231	-0.166
Mod3	-0.003	-0.273	-0.178
Sign	+0.117	+0.043	+0.015
Sign×Parity	+0.050	+0.107	+0.042
Interval	+0.503	+0.508	+0.463

The margin, stated before the celebration: the Interval pooled value is +0.503 against a registered floor of 0.500 — it cleared by 0.003, and a different seed could plausibly have missed. This is the same razor margin the loops result carries (§5), and it gets the same treatment: the Phase 1 pass, taken alone, established nothing beyond “worth retesting.” What the claim actually rests on is Phase 1b, where the fresh worlds passed with real margin. The stratified median (+0.508) and input-only variant (+0.463) cleared their 0.300 floors comfortably in all three worlds.

T0 held on all 1,479 exact-trajectory runs. One registered hand anchor was *wrong* and the validation gate caught it before any survey result was read: “increment respects parity classes” is false under saturating arithmetic ($\text{inc}(8) = 8$ stays even), a logged amendment that also surfaced an incidental mechanism — boundary saturation converts phantom into error.

Robustness (Phase 1b) [R]. Retested under a registered kill (K-R: pooled < 0.300 in the fresh world ends the predictor claim). World B ($\Sigma = \{-12..12\}$, fresh seed, fresh programs and inputs, saturation): pooled **+0.538**, stratified median +0.522 — passes the unchanged thresholds, this time with margin. World C (same but wrap-around arithmetic, registered as *adversarial* because wrapping shatters interval hulls): pooled **+0.629**, stratified median +0.618 — stronger, in the world expected to hurt. Pre-registered reading: the pass is a property of the closure’s shape, not of op-hull alignment. (Logged without weight: wrap produces more error and more spread, which mechanically flatters rank correlations; the conclusion does not depend on +0.629 exceeding +0.538.)

4. The partition side: a null with a proof

The four partition domains show no signal, and half of the reason is a proposition, not a statistic.

Proposition 4.1 (the abstract run is blind to within-class cardinality). Let ρ_π be the closure of a partition π ($\rho_\pi U$ = the union of the classes U meets), and let $M(U) = \{C \in \pi : U \cap C \neq \emptyset\}$ be the met-class profile. Then the entire stepwise-bca abstract trajectory from input U , and hence its output a_n , is a function of $M(U)$ and the operation sequence alone; and $s(U)$ is not a function of $M(U)$ — any profile containing a partially-met class C is realized by inputs of every deficiency $1 \leq |\rho U \setminus U| \cap C \leq |C| - 1$ in that class.

Proof. $a_0 = \varrho_\pi U = UM(U)$ is determined by $M(U)$; each subsequent $a_{k+1} = \varrho_\pi(f_{k+1}(a_k))$ is a function of a_k . For the second half, fix the profile and vary how many elements of a partially-met class are present. ■

The proposition delimits exactly what it delimits — it is an *information-channel impossibility result for this particular predictor*, not the familiar observation that partitions correspond to equivalence relations: cardinality-style phantom information has *no channel through the abstract computation*. It does not by itself forbid a correlation arriving through the concrete side — $\varepsilon = |a_n \setminus \varrho(V_n)|$ also depends on where the concrete trajectory lands — so whether any concrete-side channel exists is an empirical question, and that is the question the registered experiments answered: none does. Pooled correlations -0.003 to $+0.117$ across the four domains [R], and the corrected candidate — $c(V) =$ the number of partially-met classes, which unlike s is a profile-adjacent functional and was the coarsest one that could have mattered — was tested in Phase 1b with its own thresholds (≥ 0.300 pooled in ≥ 3 of 4 domains) and its own closure rule (K-P: all four below 0.200 closes the partition side, functional-shopping stops at two). **K-P fired:** pooled -0.069 (Parity), -0.099 (Mod3), $+0.055$ (Sign), $+0.008$ (Sign×Parity) [R].

The honest statement of the null, with its two grades kept separate: the abstract-side blindness is proved (Proposition 4.1); the absence of a concrete-side rescue is registered empirical fact on this toy [R]. For partition closures, error is governed by *op-class alignment* — whether the program's operations respect the partition — which is a property of the dynamics, not of the elements. Prediction there requires op-awareness, and an intrinsic per-element instrument is the wrong tool. We regard this as the second-most useful result in the note: it says which abstractions can in principle be audited statically by conflation counting (gap-filling ones) and which can only be profiled by running them (partition ones).

5. Phase 2: loops and widening

Real interval analysis loses its precision in loops, through widening, so the reviewer's question was registered and run before any write-up (PHASE2-SPEC.md): toy language extended with `while` loops over five guards, concrete collecting semantics computed exactly as a finite fixpoint, abstract semantics by the standard widening iteration (unstable bounds jump to the world bounds; no narrowing — its absence makes the world harder for the *analysis*, not for the claim).

The structural threat, hand-witnessed before the run [R]. Widening manufactures imprecision that is not intrinsic to the states, so T0 is *provably false* on loops. Registered anchor A-W: `while (x < 3) inc` on input {0} — every set the computation touches is an exact interval, intrinsic phantom $P = 0$, yet the widened invariant jumps to $[0, 8]$ and $\varepsilon = 5$. Error at zero phantom, by construction. At scale the mechanism is common: 286 of 2,400 loop runs (a quarter of all error-bearing runs) have $\varepsilon > 0$ with $P = 0$.

Result [R]. Loop corpus (50 programs, half the items loops, 2,400 runs, $\varepsilon > 0$ in 49.5% — more than double the straight-line rate): pooled Spearman(P, ε) = **+0.306** against a registered kill line of 0.300. The registered verdict band calls this *survives attenuated*; we report the margin at full strength —

0.006 of Spearman, a different seed could plausibly have fired the kill. The registered autopsy is what makes the number a mechanism rather than an ambiguity. Widening jump mass W (an analysis-side meter, registered as descriptive) predicts error at $+0.491$ — better than the intrinsic count. Conditioning on W : within the widening-quiet tercile, $q(P, \varepsilon) = +0.457$, near straight-line strength; within the widening-heavy tercile, $+0.102$ — swamped. The pooled figure is just this mixture.

One amendment, logged before any loop-corpus number was read: the continuity gate (loop-free control must reproduce the Phase 1 result) fired at $+0.470$ against a registered 0.500 — the same statistic that sat at $+0.503$ and $+0.538$ in the two earlier corpora, i.e. a quantity living *near* the threshold, hard-gated on a smaller fresh-seed corpus; an exact cross-engine check — the identical 1,440 loop-free runs through the Phase 1 engine and the new harness — returned zero mismatches in both ε and P , so the gate had mistaken sampling variation for harness error and was replaced by the strictly stronger equality check. Loop-corpus thresholds were not touched.

The qualified claim after Phase 2. Intrinsic phantom predicts interval-analysis error on straight-line code and on loop code where widening does not fire; where widening fires, the analysis's own manufactured imprecision dominates, and the quantity that predicts *that* (W) requires running the analysis — which is exactly what an intrinsic instrument exists to avoid. The qualifier is permanent absent a narrowing study [O].

6. One theorem, and the program this came from

The instrument arrived here from a program studying observer-relative structure in Heyting algebras — which distinctions survive coarse-graining (the *aperture* of an element: under how many nuclei it remains *ordinary* in Citkin's sense) and what coarse-graining destroys (the *co-aperture*: total phantom mass over all observers; closed forms on chains and divisor lattices, exact multiplicativity on products, and independence from the aperture in both directions, all kernel-checked — Brink 2026d, 2026f). Phase 0 of the present study was a literature confrontation, recorded in the spec with a dated retraction where needed: the program's *dynamical compatibility* notion (observers commuting with a time step) decomposes exactly as backward $\wedge$ forward completeness in the GRS sense — a hand proof re-checked by exhaustive oracle over 131,320 (step, observer) pairs with zero mismatches — so that layer is absorbed by existing completeness theory and is claimed by no one here [C/R].

What survived the confrontation is one closure theorem apparently outside the existing theory's predictions:

Theorem 6.1 (meet-half of the lattice conjecture) [K] (`compatible_union`, `PocketMeetHalf.lean`). On the down-set algebra of a finite poset, with observers the sub-ecosystem nuclei j_S and the time step a single-edge order extension: if S and T are both compatible with the step (equivalently, both backward- and forward-complete for it), then $S \cup T$ is compatible. Since $j_{\{S \cup T\}}$ is the pointwise meet of j_S and j_T — the reduced product of the two domains — this is a *completeness-preservation result for the reduced product of tied domains*. To forestall an over-reading: the reduced product and the

completeness theory of domain combinations are classical (Cousot–Cousot 1979; Giacobazzi–Ranzato and successors); what is claimed is a new theorem in this specific algebraic setting — these observers, this step — not a new reduced-product principle.

Why the theorem is not forced [K] (step_not_meet_preserving). There is a general route by which such closures come for free: if the step preserved binary meets, meet-closure of the compatibles would follow formally. That route is provably unavailable here. The step fails binary meet-preservation, with a kernel-checked two-element witness: for incomparable a, b , take $U = \downarrow a$ and $V = \downarrow b$; the element a lies in $\text{step}(U) \cap \text{step}(V)$ — the step adds $\downarrow a$ to V because V reaches b — but not in $\text{step}(U \cap V)$, which never reaches b . (This is the kernel form of the universal witness recorded in the pocket-study spec, Amendment 3.) The proof of `compatible_union` accordingly cannot and does not factor through any preservation property of the dynamics; its mixed case — b visible to one observer’s image but not the other’s — is closed by the key lemma `mem_obs_of_le_a`: compatibility forces the whole cone $\downarrow a$ into any image that reaches b . The trajectory corollary (compatibility with each step composes over the whole trajectory, `survival_composes`) sits in the same file.

Whether Theorem 6.1’s mechanism extends beyond Alexandrov carriers and single-edge steps is open [O] and is the one theoretical thread this note leaves deliberately hanging.

7. What this note claims, and what it does not

Claimed. The instrument and its confusion-class semantics at closure-operator generality [K]; the exact bridge $s = \delta(U, qU)$ to the partial-completeness framework [C]; the three registered experimental results with their kill lines — the gap-filling pass with three-world robustness, the partition null (Proposition 4.1 for the abstract-side blindness, registered experiment for the absence of a concrete-side rescue), the widening attenuation with its measured margin and autopsy [R]; the meet-half theorem, together with the kernel-checked witness that it is not forced by any meet-preservation property of the step [K].

Not claimed. The count, as a functional — it is the POPL 2022 quasi-metric at (U, qU) and the MCAI false-positive count at $q = \gamma \circ \alpha$, and this note would be mispositioned as “a new measure of imprecision”; the territory of measuring realized imprecision is occupied by the quasi-metric, pre-metric, and model-counting families cited in §1, and what is claimed is the *prior, predictive* direction with its map. That any of this holds on infinite carriers, relational domains (octagons, polyhedra), or real analyzers — the worlds here are finite toys computed exactly, and the distance to practice is the note’s largest caveat, stated without decoration. That phantom mass improves on ε -partial completeness at its own job (it does a different job: prior and per-element rather than post-hoc and per-run; the two are complementary by construction). That the four-position/ordinariness layer of the source program transfers to abstract domains — that requires non-Boolean carriers (Giacobazzi–Scozzari’s Heyting completion is the established route) and is untouched here [O]. That the widening result generalizes to narrowing regimes [O].

The methodological claim we stand behind. Every experiment was registered with thresholds and kill conditions before its code existed; two registered expectations were wrong (a hand anchor

falsified by the world’s own boundary convention; a continuity gate that mistook sampling variation for harness error) and both amendments are dated, logged with their nature, and were made before the gated results were read; one kill fired (K-P) and closed its side of the study permanently; one result survived by 0.006 and is reported at that margin. A quantitative claim gains more from a mechanism that can refuse than from a threshold it happens to clear.

8. Formalization and artifacts

All formal results in Lean 4 against Mathlib4; no `sorry`, no `native_decide`; axiom audits at most `propext`, `Classical.choice`, `Quot.sound`.

Result	Lean theorem	File
Confusion class at uco generality (2.2)	<code>ClosureOperator.le_apply_iff</code>	<code>Lattice/CoApertureClosedForm.lean</code>
Confusion class, nucleus form (special case)	<code>IsNucleus.le_apply_iff</code>	ibid.
Meet-half theorem (6.1)	<code>compatible_union</code>	<code>Lattice/PocketMeetHalf.lean</code>
6.1 not forced: step fails meet-preservation	<code>step_not_meet_preserving</code>	ibid.
Key lemma for 6.1	<code>mem_obs_of_le_a</code>	ibid.
Trajectory composition	<code>survival_composes</code>	ibid.
Co-aperture closed forms, multiplicativity, independence	<code>CoApertureClosedForm.lean</code>	<code>Lattice/CoApertureClosedForm.lean</code>
Alexandrov Booleanization (context for §6)	<code>card_regular_eq_two_pow</code>	<code>Lattice/AlexandrovBoolean.lean</code>

Proposition 4.1 is proved inline in §4 (four lines) and is not kernel-checked; it carries no [K] tag.

Empirical artifacts, all with dated registered specs, amendments, and postscripts: `absint-dictionary/SPEC.md` (Phase 0 literature confrontation, including the recorded retraction), `PHASE1-SPEC.md`, `PHASE1B-SPEC.md`, `PHASE2-SPEC.md`, scripts `01-worked-example.py`, `02-phase1b.py`, `03-phase2-loops.py`, raw outputs under `absint-dictionary/out/`. Oracle re-check of the decomposition and reduced-product hand proofs: `pocket-study/06-decomposition-check.py`. Source: github.com/thefalsework/papers.

Disclosure. Drafting and formalization were AI-assisted under direction, per the project’s validation architecture and its framework for epistemic dependency (Brink 2026e); the grade table above is the author’s warrant.

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